Prog7312 ICE Task 1

Project: IoT Temperature Controller Simulation

The **Internet of Things (IoT)** refers to a network of physical devices that are connected to the internet and can collect, exchange, and process data. Examples of IoT devices include smart thermostats, temperature sensors, smart lighting systems, security cameras, and environmental monitoring systems. These devices continuously monitor conditions such as temperature, humidity, and motion, allowing users to make informed decisions or automate processes.

In this project, you must develop a **Windows Forms Application** that simulates an **IoT Temperature Controller**. The application represents a simple smart temperature monitoring system where a user can manually adjust the temperature and observe how the sensor value changes over time. The application must display the current temperature while maintaining a history of all temperature adjustments. This simulates how real IoT systems record sensor readings and maintain logs for monitoring, troubleshooting, and data analysis.

Functional Requirements

Develop a C# Windows Forms (.NET Framework) application that simulates an IoT Temperature Controller. Your application must meet the following requirements:

1. Implement a temperature controller that initializes with a default temperature of 25°C and displays the current temperature when the application starts.
2. Provide functionality for users to increase and decrease the temperature. Each adjustment must update the current temperature display and record the change in a temperature history log.
3. Implement a Reset feature that restores the temperature to its default value of 25°C, updates the current temperature display, and records the reset action in the history log.
4. Include a Clear History function that removes all entries from the temperature history without affecting the current temperature.
5. Implement an Exit function that displays a confirmation dialog before closing the application.
6. Ensure the user interface provides real-time feedback by displaying the current temperature and maintaining a history of all temperature adjustments throughout the application's execution.

Submission

1. **Provide a short and clear report describing the data structures used in the project and justify why each data structure is the most appropriate for its intended purpose.**
2. **Provide screenshots (windows form and output) of every stage/task**
3. **Provide a complete project code**
4. **Submission should be made Arc, under ICE Task 1**